Total number of printed pages–4

4R/2025 PHY4304C

2025

PHYSICS

(Major–III)

Paper : PHY4304C

(Ancient Indian Astronomy)

Full Marks : 45

Time : Two hours

The figures in the margin indicate full marks for the questions.

1. Answer the following questions : 1×4=4

(a) What is a great circle ?

(b) What is *madhyama-graha* (मध्यम-ग्रहः) ?

(c) Who wrote the 'Vedanga Jyotisham' (वेदाङ्गज्योतिषम्) ?

(d) What is the meaning of Ayanachalana (अयनचलनः) ?

2. Answer the following questions : $2 \times 3 = 6$

(a) State the assumptions made by Aryabhatta while deriving the *Sighra* (शीघ्र) daily motion.

(b) What is the difference between the interpretations of comets according to the Vedas and the *Sūrya Siddhānta* (सूर्य-सिद्धान्त) ?

(c) Enumerate the astronomical classifications recognized during the Pre-Siddhantic Period.

3. Answer the following questions as directed [*(a)* is **compulsory**. Answer ***either*** *(b)* **or** *(c)* and ***either*** *(d)* **or** *(e)*]

(a) Use epicyclic model of indian planetary system to find out the expression of Longitude of a planet. 5

(b) Write a short note on Chāyā Shankū (छाया शंकु). 5

OR

(c) Write a short note on *Ghatika Patra* (घटिकापात्र)

(d) Write a short note on time-keeping during vedic period of astronomy particularly mentioned in *Yajurveda*. 5

OR

(e) Write briefly about the Vedic Rituals performed for astronomy.

4. Answer the following questions as directed :

(a) Find the expression of the daily motion (diurnal motion) of a planet at *manda* (मन्द) as laid down in *Sūrya Siddhānta* (सूर्य-सिद्धान्तः) 10

OR

(b) Derive the expression of the daily motion (diurnal motion) of a planet at *sighra* (शिघ्र) as mentioned in *Sūrya Siddhānta* (सूर्य-सिद्धान्तः)

5. Answer the following questions as directed :

(a) *Sūrya Siddhānta* (सूर्य-सिद्धान्तः) explains lunar eclipses as the result of the earth's shadow (specifically the umbra) falling on the moon when it comes directly opposite to the sun using geometric and trigonometric principles. Based on this find out the following : 3+4+3=10

(i) Derive of the angular diameter of Earth's Shadow at the Moon's Distance.

(ii) Calculate of Half Duration of a Lunar Eclipse.

(iii) Write the formula for Eclipse magnitude and hence show the various eclipse conditions.

OR

(b) Calculate the *ahargana* (अहर्गण) for 30 May, 2025 and hence calculate the *tithi* (तिथि) for that date as well. 7+3=10